Serious adverse events after cessation of nucleos(t)ide analogues in individuals with chronic hepatitis B: A systematic review and meta-analysis

Cheng-Hao Tseng, Tzu-Haw Chen, Jia-Ling Wu, Teng-Yu Lee, John A. Borghi, Jaw-Town Lin, Mindie H. Nguyen, Yao-Chun Hsu

Table of contents

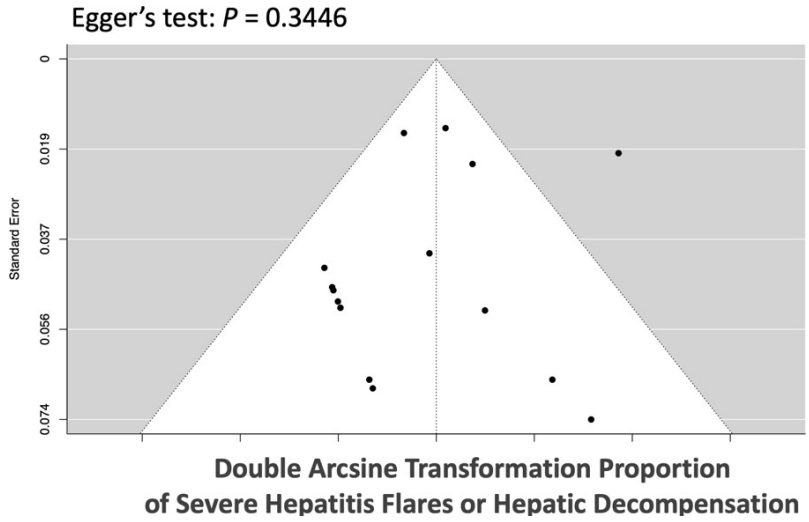
Supplementary figures.....2

Supplementary tables.....11

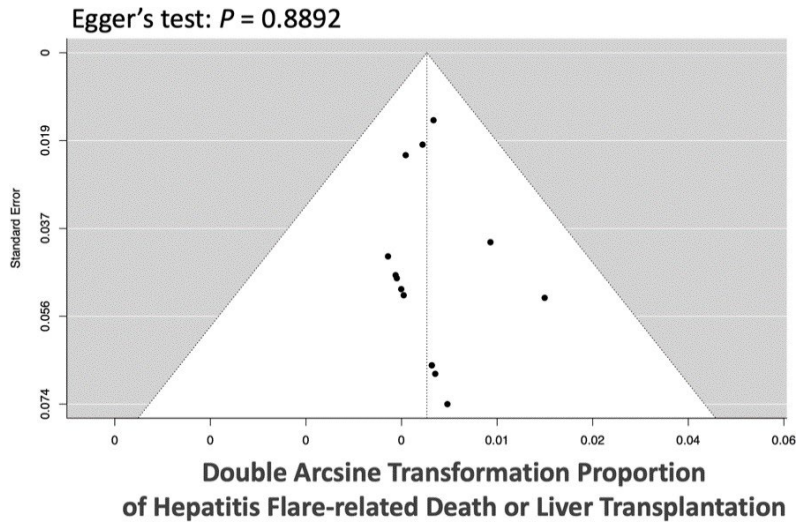
Supplementary references.....46

Fig. S1: Funnel Plot of the 15 Studies for risk estimation[a]

A: Selected Studies for Severe Hepatitis Flares or Hepatic decompensation



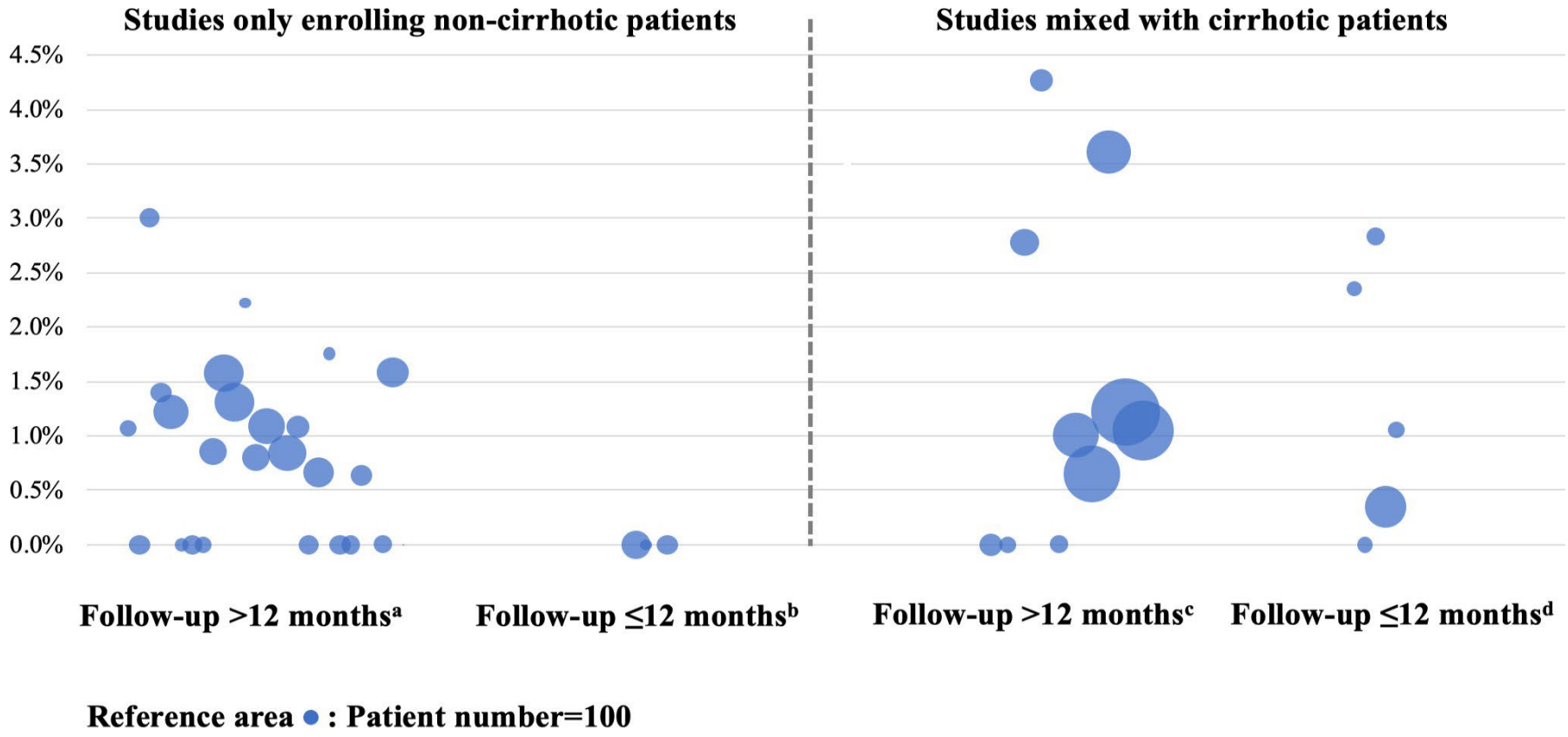
B: Selected Studies for Hepatitis Flare-related Death or Liver Transplantation



a: criteria of study selection for pooling aggregate data: (1) no overlapping populations, (2) follow-up duration >12 months, (3) not for special populations such as studies exclusively composed of patients with prior severe acute exacerbation or low serum HBsAg level.

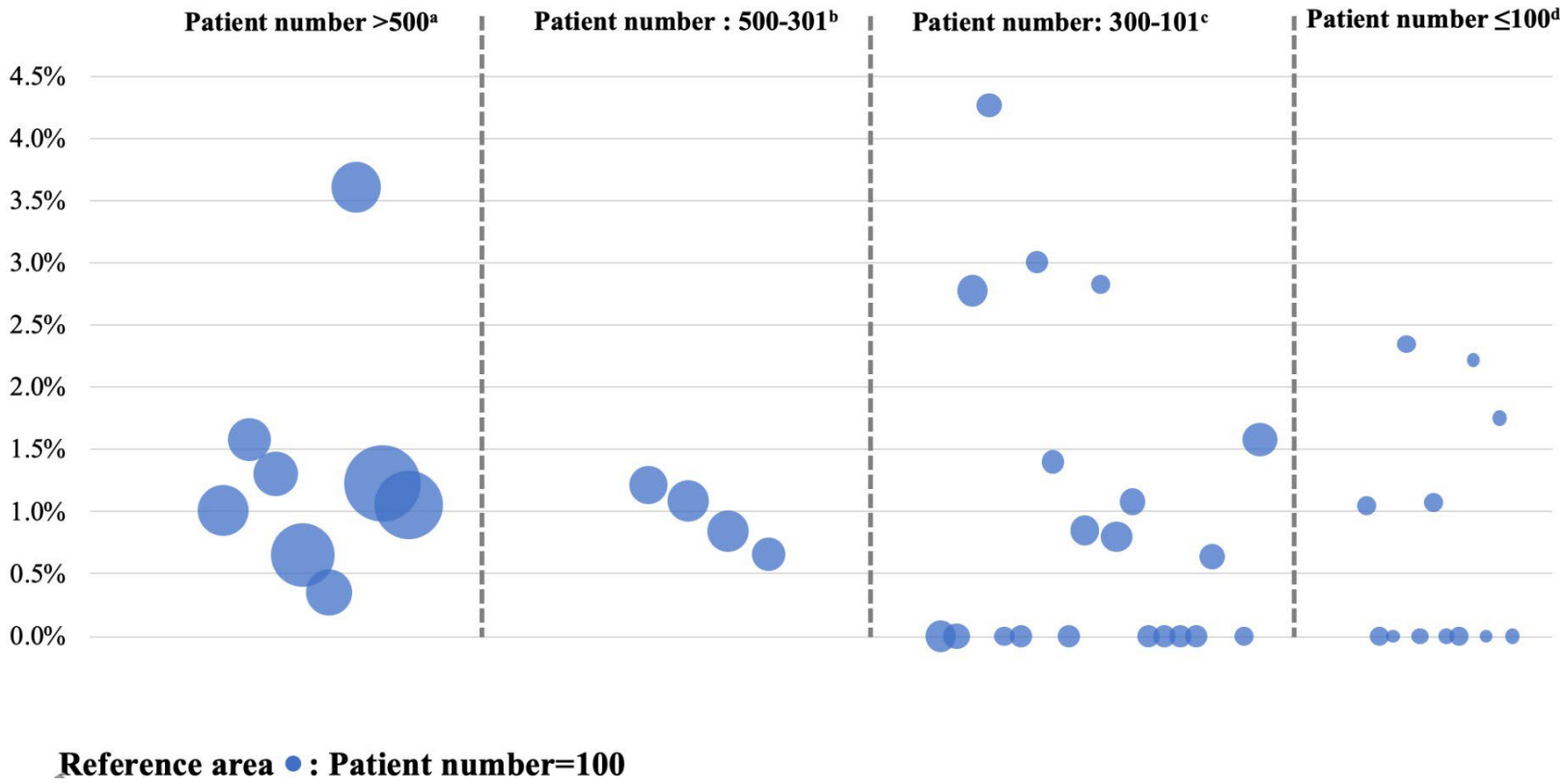
HBsAg: hepatitis B virus surface antigen

Fig. S2
Proportions of Severe Hepatitis Flares or Hepatic Decompensation



Each circle indicates a study and its area represents the sample size of the study.
a: reference[1-24], b: reference[25-27], c: reference[28-37], d: reference[38-42]

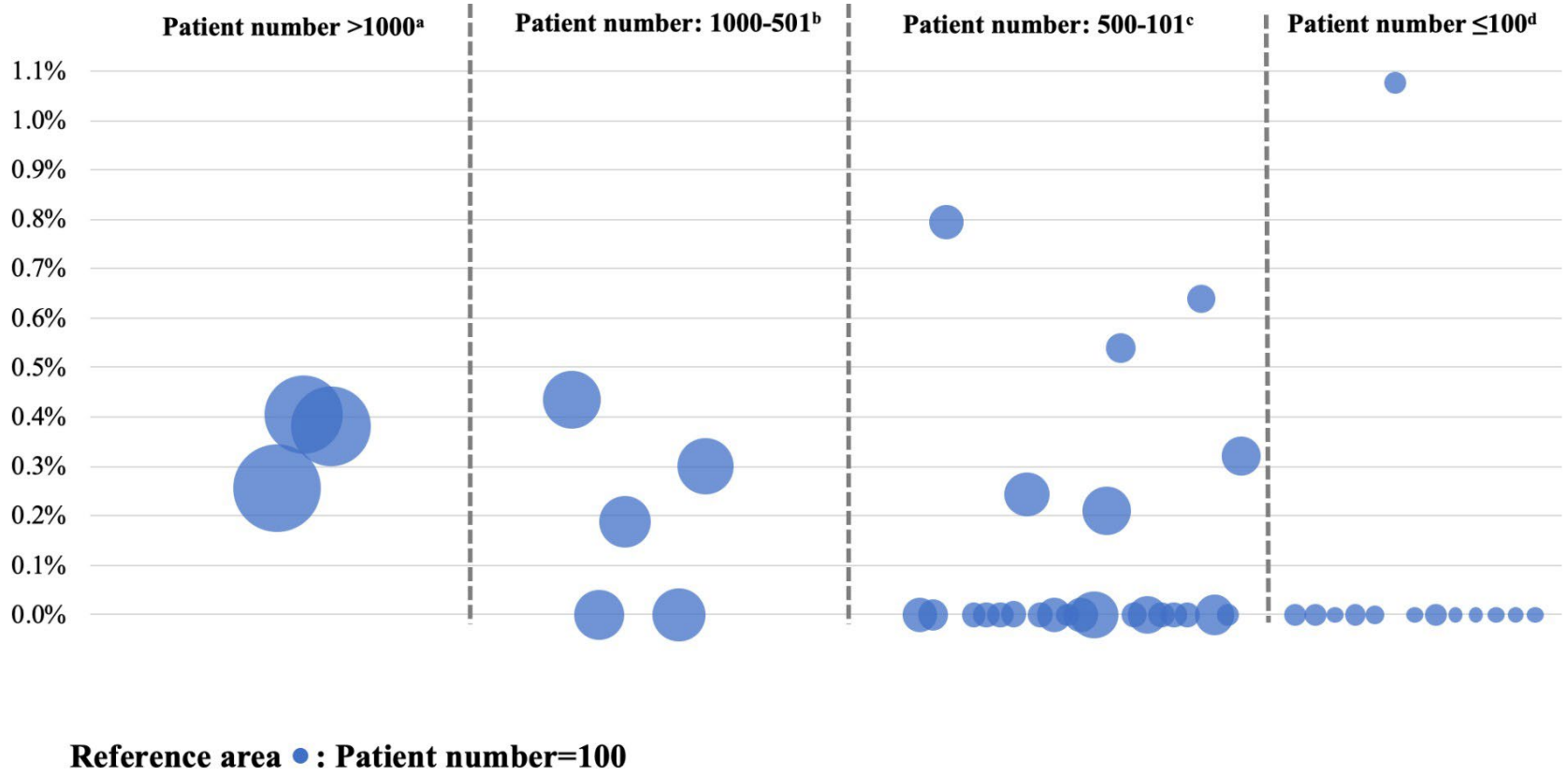
Fig. S3
Proportions of Severe Hepatitis Flares or Hepatic Decompensation by Sample Size



Each circle indicates a study and its area represents the sample size of the study.
a: reference[9, 10, 33-37, 41], b: reference[13, 14, 17, 43], c: reference[2-4, 6, 8, 12, 15, 16, 19-21, 23-25, 27, 28, 30-32, 40], d: reference[5, 7, 11, 18, 22, 26, 29, 38, 39, 42, 44]

Fig. S4

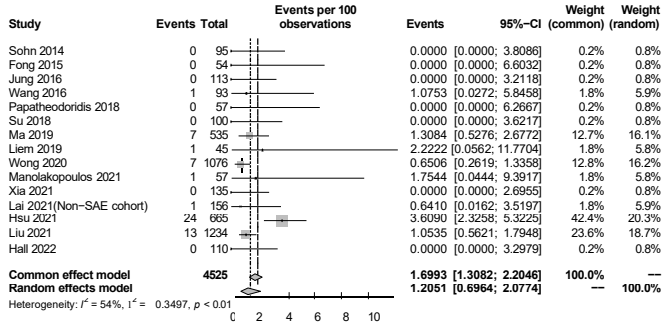
Proportions of Hepatitis Flare-related Liver Death, or Liver Transplantation by Sample Size



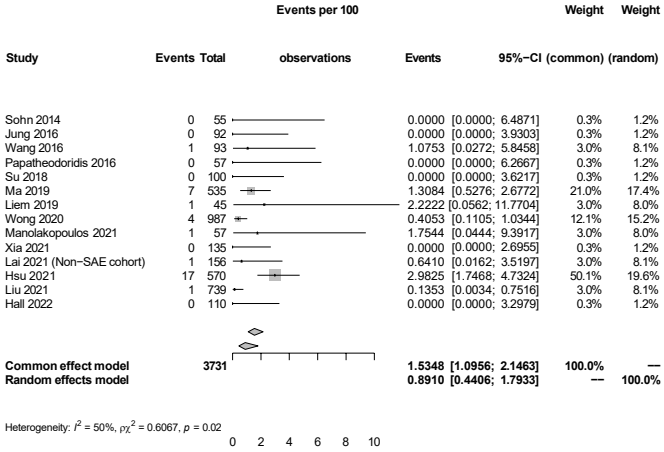
Each circle indicates a study and its area represents the sample size of the study.
a: reference[36, 37, 45], b: reference[9, 10, 33, 35, 41], c: reference[2-4, 6, 12-17, 19-21, 23-25, 27, 28, 30, 32, 40, 43, 46], d: reference[1, 5, 7, 11, 18, 22, 26, 29, 38, 39, 42, 44, 47]

Fig. S5: Pooled Proportions for Severe Hepatitis Flares or Decompensation from Selected Studies

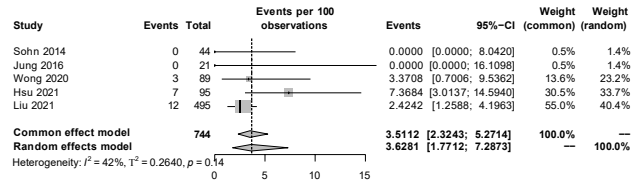
5A: Overall Populations



5B: Noncirrhotic Populations



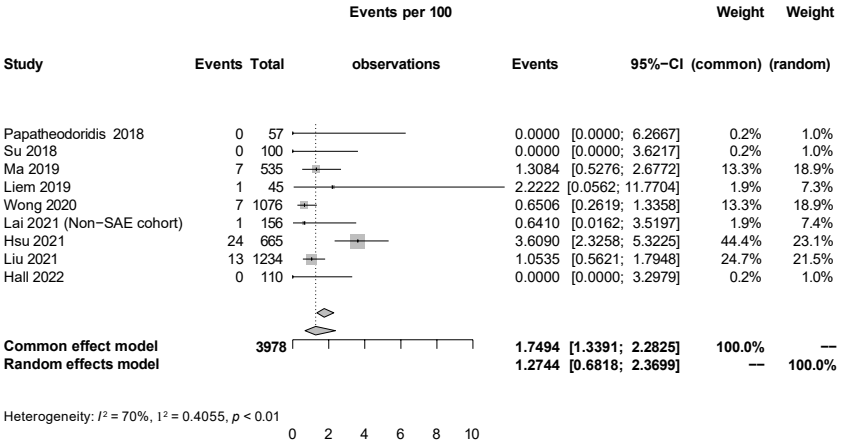
5C: Cirrhotic Populations



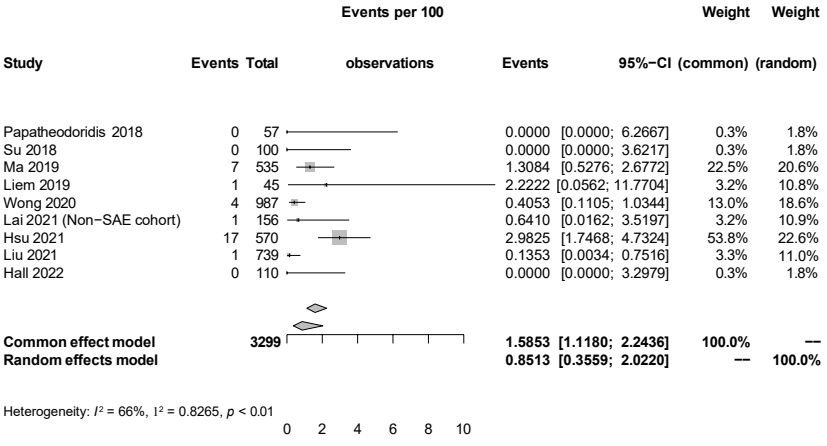
Criteria of study selection for pooling aggregate data: (1) no overlapping populations, (2) follow-up duration >12 months, (3) not for special populations such as studies exclusively composed of patients with prior severe acute exacerbation or low serum HBsAg level. Estimated risks, generated by random-effects model and the inverse variance method, are expressed by proportions with 95% confidence internal. Between-study heterogeneity are represented by I^2 statistics. HBsAg: hepatitis B virus antigen, Non-SAE cohort: patients without severe acute exacerbation (bilirubin > 2mg/dL or prolonged prothrombin time >3 seconds) before antiviral agents

Fig. S6: Pooled Proportions for Severe Hepatitis Flares or Decompensation from Selected Studies with Clear Definitions

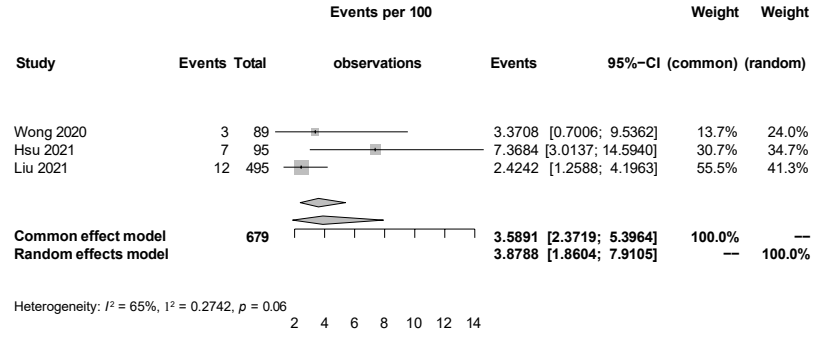
6A: Overall Populations



6B: Non-cirrhotic Populations



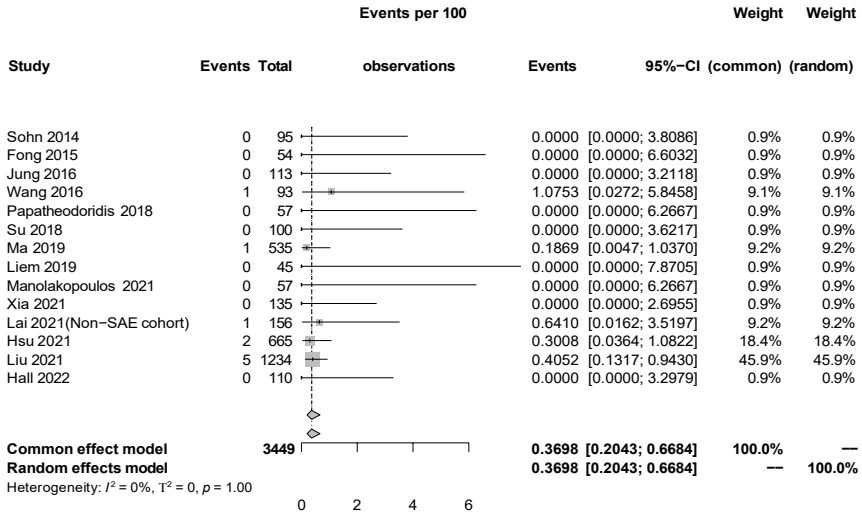
6C: Cirrhotic Populations



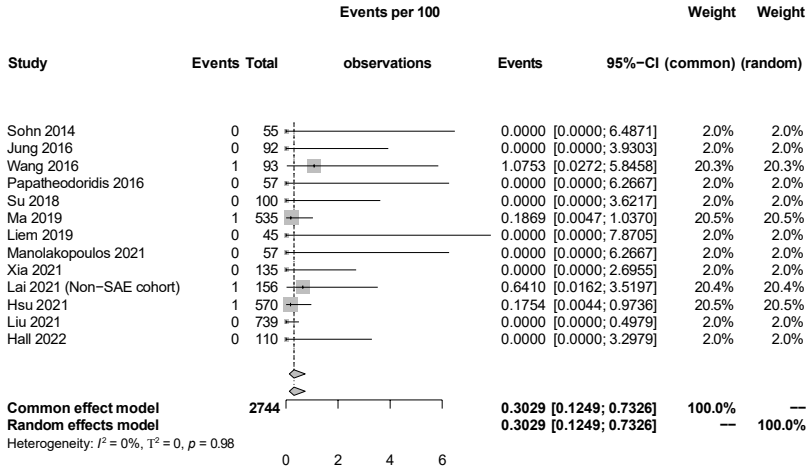
Criteria of study selection for pooling aggregate data: (1) no overlapping populations, (2) follow-up duration >12 months, (3) not for special populations such as studies exclusively composed of patients with prior severe acute exacerbation or low serum HBsAg level. Estimated risks, generated by random-effects model and the inverse variance method, are expressed by proportions with 95% confidence internal. Between-study heterogeneity are represented by I^2 statistics. HBsAg: hepatitis B virus surface antigen, Non-SAE cohort: patients without severe acute exacerbation (bilirubin > 2mg/dL or prolonged prothrombin time >3 seconds) before antiviral agents

Fig. S7: Pooled Proportions for Hepatitis Flare-related Death, or Liver Transplantation from Selected Studies

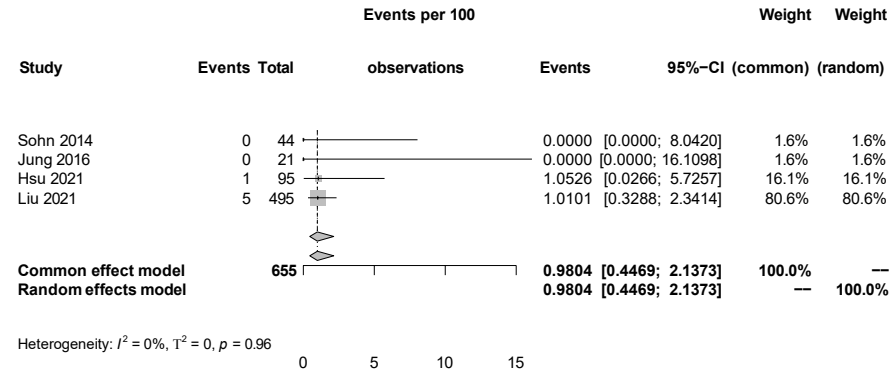
7A: Overall Populations



7B: Non-cirrhotic Populations



7C: Cirrhotic Populations



Criteria of study selection for pooling aggregate data: (1) no overlapping populations, (2) follow-up duration >12 months, (3) not for special populations such as studies exclusively composed of patients with prior severe acute exacerbation or low serum HBsAg level. Estimated risks, generated by random-effects model and the inverse variance method, are expressed by proportions with 95% confidence internal. Between-study heterogeneity are represented by I^2 statistics. HBsAg: hepatitis B virus surface antigen, Non-SAE cohort: patients without severe acute exacerbation (bilirubin > 2mg/dL or prolonged prothrombin time >3 seconds) before antiviral agents.

Table S1: PRISMA Check List

Section	Item #	Checklist item	Location
TITLE			
Title	1	Identify the report as a systematic review.	Page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Page 5-6
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page 8-9
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 9
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Page 9-10
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page 9
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Page 9
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page 9
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 9-10
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page 9-11
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page 9-11
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Page 12
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Page 12
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item	Page 12

Section	Item #	Checklist item	Location
		#5)).	
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	NA
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	NA
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Page 12
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Page 12
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	NA
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Page 12
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	NA
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page 12-13
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Page 12-13
Study characteristics	17	Cite each included study and present its characteristics.	Page 13-14
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Page 13
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Page 13-18
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Page 18-19
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Page 18-19
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	NA
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Page 13
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	NA
DISCUSSION			

Section	Item #	Checklist item	Location
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Page 19-24
	23b	Discuss any limitations of the evidence included in the review.	Page 24-25
	23c	Discuss any limitations of the review processes used.	NA
	23d	Discuss implications of the results for practice, policy, and future research.	Page 25
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	NA
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	NA
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 3
Competing interests	26	Declare any competing interests of review authors.	Page 3
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	NA

Table S2

Pubmed 2006.01.01-2022.08.18

#7	((((#1) AND (#2)) AND (#3)) NOT (#4)) NOT (#5) Filters: from 2006/1/1 - 2022/8/18	**1,459**
#6	((((#1) AND (#2)) AND (#3)) NOT (#4)) NOT (#5)	**1,960**
#5	("Editorial"[PT] OR "Letter"[PT] OR "Case Reports"[PT] OR "Review"[PT] OR "Systematic Review"[PT] OR "meta-analysis"[PT])	**6,873,435**
#4	("Child"[MH] OR "Infant"[MH] OR Child[TW] OR Children[TW] OR Infant[TW] OR Infants[TW])	**3,134,472**
#3	("Withholding Treatment"[MH] OR Discontinue[TW] OR Discontinuing[TW] OR Discontinued[TW] OR Discontinuation[TW] OR Stop[TW] OR Stopping[TW] OR Stopped[TW] OR Cessation[TW] OR "Finite Treatment"[TW] OR End[TW] OR Ending[TW] OR "Off treatment"[TW] OR "Withholding Treatment"[TW] OR "Drug withdrawal"[TW])	**1,183,469**
#2	("Antiviral Agents"[MH] OR "Antivirus agent"[TW] OR "Antivirus agents"[TW] OR "Antiviral Agent"[TW] OR "Antiviral Agents"[TW] OR Antivirals[TW] OR Nucleoside[TW] OR Nucleotide[TW] OR Nucleostide[TW] OR Entecavir[TW] OR Tenofovir[TW])	**637,747**
#1	("Hepatitis B, Chronic"[MH] OR "Hepatitis B virus"[MH] OR "Hepatitis B"[TW])	**107,968**

EMBASE 2006.01.01-2022.08.18

#6	#5 AND ([article]/lim OR [article in press]/lim) AND [1-1-2006]/sd NOT [18-08-2022]/sd	3,228
#5	#1 AND #2 AND #3 NOT #4	8,122
#4	'child'/exp OR 'infant'/exp OR child:ti,ab,kw OR children:ti,ab,kw OR infant:ti,ab,kw OR infants:ti,ab,kw	3,753,820
#3	'drug withdrawal'/exp OR discontinue:ti,ab,kw OR discontinuing:ti,ab,kw OR discontinued:ti,ab,kw OR discontinuation:ti,ab,kw OR stop:ti,ab,kw OR stopping:ti,ab,kw OR stopped:ti,ab,kw OR stoppage:ti,ab,kw OR cessation:ti,ab,kw OR 'finite treatment':ti,ab,kw OR end:ti,ab,kw OR ending:ti,ab,kw OR 'off treatment':ti,ab,kw OR 'withholding treatment':ti,ab,kw OR 'drug withdrawal':ti,ab,kw	1,804,671
#2	'antivirus agent'/exp OR 'antivirus agent':ti,ab,kw OR 'antivirus agents':ti,ab,kw OR 'antiviral agent':ti,ab,kw OR 'antiviral agents':ti,ab,kw OR antivirals:ti,ab,kw OR nucleoside:ti,ab,kw OR nucleotide:ti,ab,kw OR nucleostide:ti,ab,kw OR entecavir:ti,ab,kw OR tenofovir:ti,ab,kw	1,703,696
#1	'hepatitis b'/exp OR 'hepatitis b':ti,ab,kw	156,653

WOS: 2006.01.01-2022.08.18

#8	#5 NOT #4 and Review Articles or Meeting Abstracts or Editorial Materials or Letters or Book Chapters or Corrections or Reprints or Publication With Expression Of Concern or Notes (Exclude – Document Types) and 2022 or 2021 or 2020 or 2019 or 2018 or 2017 or 2016 or 2015 or 2014 or 2013 or 2012 or 2011 or 2010 or 2009 or 2008 or 2007 or 2006 (Publication Years)	1,610
#7	#5 NOT #4 and Review Articles or Meeting Abstracts or Editorial Materials or Letters or Book Chapters or Corrections or Reprints or Publication With Expression Of Concern or Notes (Exclude – Document Types)	2,042
#6	#5 NOT #4	2,502

#5	#1 AND #2 AND #3	2,687
#4	ALL=(child OR children OR infant OR infants)	2,531,507
#3	ALL=(discontinuing OR discontinued OR discontinuation OR stop OR stopping OR stopped OR stoppage OR Cessation OR finite treatment OR end OR ending OR off treatment OR withholding treatment OR drug withdraw)	1,912,875
#2	ALL=(antivirus agents OR antivirus agent OR antiviral agents OR antiviral agent OR antivirals OR entecavir OR tenofovir OR nucleoside OR nucleotide OR nucleostide)	526,999
#1	Hepatitis B (All Fields)	148,644

Table S3: Risk of Bias Assessment Form

NEWCASTLE - OTTAWA Risk of BIAS ASSESSMENT SCALE for COHORT STUDIES

Note: A study can be awarded a maximum of one star for each numbered item within the Selection and Outcome categories. A maximum of two stars can be given for Comparability

Selection Score: ( ) (Maxium:4)

1) Representativeness of the Off-NA cohort (I)
   a) Stopping rule according to or more strict than international suggestion) *
   b) Stopping rule according to local regulation*
   c) No clearly mentioned the stopping rule or criteria loose than (a) or (b)
   d) No mentioned of stopping rule

3) Representativeness of the Off-NA cohort (II): for avarage population
   a) yes (Definition: **<50%** patients with cirrhosis; **Not** specific for population lower HBsAg leve: <200 IU/ml)l *
   b) no

2) Ascertainment of off-NA
   a) secure record (eg surgical records) *
   b) structured interview *
   c) written self report
   d) no description

4) Case number more than 100
   a) Yes *
   b) No

Comparability Score: ( ) Max:2

1) Comparability of cohorts on the basis of the design or analysis
   a) study controls for age, gender, and cirrhosis status **
   b) study controls for any additional factor other than age, gender, or cirrhosis status*
   c) No control

Outcome Score: ( ) Max:3

1) Assessment of outcome
   a) independent blind assessment *
   b) record linkage *
   c) self report
   d) no description

2) Was follow-up long enough for outcomes to occur
   a) mean or median follow-up longer than 1 years *
   b) no

3) Designe for assessment of clinica outcome
   a) Sever flare events were prespecified as study outcome *
   b) Mention sever flare events, but not prespecified as study outcome
   d) no statement

Table S4: Results of Risk of Bias Assessment by NEWCASTLE-OTTAWA Scale

	Selection	Comparability	Outcome	Total Score
	0-4	0-2	0-3	
Chen[16] 2021	4	0	2	6
Fang[14] 2021	4	0	2	6
Huang[17] 2021	4	0	2	6
Shouval[25] 2009	4	0	2	6
Jeng[42] 2013	3	0	2	5
Seto[28] 2015	4	0	2	6
Sohn[29] 2014	3	0	2	5
Chen[30] 2015	4	0	2	6
Chen[31] 2015	3	0	2	5
Fong[44] 2015	1	0	2	3
Jeng[38] 2016	3	0	1	4
Jung[32] 2016	4	0	1	5
Lin[39] 2016	3	0	2	5
Wang[1] 2016	3	0	2	5
Chien[2] 2017	4	0	2	6
Hsu[3] 2017	4	0	2	6
Yao[48] 2017	4	0	2	6
Chen[4]	4	0	2	6

2018				
Chen[43] 2018	4	0	2	6
Jeng[33] 2018	4	0	2	6
Papatheodoridis[6] 2018	4	0	2	6
Papatheodoridis[5] 2018	3	0	2	5
Su[7] 2018	4	0	2	6
Buti[40] 2019	4	0	2	6
Chen[8] 2019	4	0	2	6
Kuo[9] 2019	4	0	2	6
Liem[11] 2019	3	2	1	6
Ma[10] 2019	4	0	2	6
Chen[12] 2020	4	0	2	6
Chiu[13] 2020	4	0	2	6
Sonneveld[41] 2022	4	0	2	6
Tseng[49] 2020	4	0	2	6
Wang[26] 2020	2	0	2	4
Wong[34] 2020	3	0	3	6
Kuo[15] 2021	4	0	2	6
Manolakopoulos[18] 2021	3	0	2	5
Seto[50] 2021	4	0	2	6

Xie[27] 2021	4	0	2	6
Xia[19] 2021	4	0	2	6
Liao[20] 2021	4	0	2	6
Lai[21] 2021	4	0	3	7
Hsu[35] 2022	4	0	3	7
Hirode[36] 2022	4	0	3	7
Papatheodoridi[46] 2022	4	0	2	6
Liu[37] 2022	4	0	3	7
Broquetas[47] 2022	3	0	1	4
Hall[24] 2022	4	0	3	7
Chen[23] 2022	4	0	2	6
Peng[45] 2022	4	0	2	6
Papatheodoridi[22] 2022	3	0	2	5

Table S5: Summary of the Demographic Data of the 50 Enrolled Studies

Author Year	Study scale	Study setting	Number	ETV and/or TDF proportion	Asian / Caucasian / others	Age	Male	Cirrhosis	Pretreatment HBeAg (+)	EOT-HBsAg >100 / >10,00 IU/mL	Nuc duration (m)	Follow-up (m)
Shouval[25] 2009	Multicenter International	Prospective	257	100%	NA	NA	NA	0%	0%	NA / NA	12(fix)	6 (fix)
Buti[40] 2019	Multicenter International	Prospective	106[f]	100%	38.71%/ NA/NA[g]	50.00[b,d,g]	78.23%[g]	25.81%[g]	33.07%	NA / NA	≥96	6 (fix)
Liem[11] 2019	Single center	RCT	45[a]	100%	98% / 2% / 0%	49.00 [b]	57.78%	0%	40.00%	92.54% / NA	91.20[b]	18 (fix)
Wong[34] 2020	Multicenter Single country	EHR	1076	95.82%	100% / 0% / 0%	59.10 [b]	74.81%	8.27%	0%	NA / NA	82.30 [b]	44.30[b]
Seto[50] 2021	Single center	Prospective	114 (All with HBsAg < 200 IU/mL)	100%	100% / 0% / 0%	58.40[c]	65.79%	0%	16.67%	22.81% / NA	80.40[c]	12 (max)
Lai[21] 2021	Single center	Retrospective	SAE group: 78	52.56%	100% / 0% / 0%	42.00[c]	71.80%	0%	32.05%	NA / NA	36.00[c]	19.56[c]
			Non SAE group: 156	60.26%		43.00[c]	71.80%	0%	39.74%	NA / NA	36.00[c]	24.87[c]
Hsu[35] 2022	Single center	EHR	665[h]	100%[h]	100% / 0% / 0%	50.30 [c,h]	73.99%[h]	14.29%[h]	26.02%	NA / NA	35.40 [c,h]	39.60[c,h]
Hirode[36]	2022	Multicenter International										

| | | |

Liu[37] 2022	Single center	Prospective	1234	100%	100% / 0% / 0%	56.39[b,d]	83.55%	40.11%	0%	82.66% / NA	35.88[c]	50.25[c]
Seto[28] 2015	Multicenter Single country	Prospective	184	100%	100% / 0%/ 0%	56.96[b,d]	67.94%	18.48%	0%	94.57% / 45.65%	36.72[b]	7.60[c,d]
Chen[30] 2015	Single center	Retrospective	252	100%	100% / 0% / 0%	46.00[b]	80.16%	13.89%	32.94%	NA / 25.79%	41.00[b]	22.50[c]
Chen[31] 2015	Multicenter Single country	Prospective and Retrospective	164	100%	100% / 0% /0%	53.40[b]	84.15%	100%	0%	NA / NA	30.30[b]	25.50[b]
Jeng[38] 2016	Multicenter Single country	Prospective and Retrospective	85	100%	100% / 0% / 0%	52.00[b]	85.88%	40.00%	0%	87.05% / 56.47%	39.25[c]	9.75[c]
Chien[2] 2017	Multicenter Single country	Prospective	140	95.71%	100% / 0% 0%	49.10[c]	77.86%	0%	27.86	NA / NA	36.60[c]	19.9[c]
Hsu[3] 2017	Multicenter Single country	Prospective	133	100%	100% / 0% 0%	NA	NA	0%	NA	NA / NA	Around 36	12.6[b]
Yao[48] 2017	Single center	Retrospective	119 (All with HBsAg ≤200 IU/mL)	71.43%	100% / 0% / 0%	52.00[b]	78.99%	23.53%	0%	42.86% / NA	37.75[b]	34.64[c,d]
Chen[4] 2018	Multicenter Single country	Retrospective	143	100%	100% / 0% 0%	47.12[b]	81.12%	0%	27.27	NA / NA	39.77[b]	13.63[c,d]
Chen[43] 2018	Single center	Retrospective	411	73.24%	100% / 0% / 0%	45.10[b]	77.86%	0%	36.01%	NA / 38.69%	37.30[b]	32.79[c,d]
Jeng[33] 2018	Single center	Prospective and Retrospective	691	92.76%	100% / 0% 0%	55.55[c]	85.96%	44.57%	0%	83.50% / 43.85%	39.00[b]	38.78[c]

Papatheodoridis[5] 2018	Multicenter Single country	Prospective	57	100%	0% / 100% / 0%	65.30[c,d]	64.91%	0%	0%	82.46% / 40.35%	63.60[b,i]	18[c]
Su[7] 2018	Single center	Prospective and Retrospective	100	100%	100% / 0% 0%	54.08[c,d]	72.00%	0%	28%	NA / NA	37.00[b]	32.35[c,d]
Chen[8] 2019	Single center[j]	Retrospective	234	100%	100% / 0% 0%	53.4[b]	80.34%	0%	0%	NA / NA	42.23[b]	52.43[c,d]
Kuo[9] 2019	Single center	Prospective and Retrospective	507	100%	100% / 0% / 0%	47.28[b]	78.11%	0%	30.37%	NA / NA	41.17[b]	23.82[c,d]
Ma[10] 2019	Single center	Prospective and Retrospective	535	100%	100% / 0% / 0%	50.71[b]	77.57%	0%	29.91%	NA / NA	41.09[b]	26.89[c,d]
Chen[12] 2020	Single center[j]	Prospective and Retrospective	250	100%	100% / 0% 0%	53.80[b]	79.60%	0%	0%	NA / NA	42.10[b]	58.10[b]
Chiu[13] 2020	Single center	Retrospective	460	100%	100% / 0% 0%	51.57[b]	80.43%	0%	0%	74.78% / NA	40.79[b]	20.01[c,d]
Tseng[49] 2020	Single center[j]	Retrospective	135 (All with HBsAg <100 IU/mL)	100%	100% / 0% / 0%	52.60[b]	77.04%	0%	13.33%	0% / 0%	42.00[b]	21.75[c]
Fang[14] 2021	Single center	Retrospective	474	100%	100% / 0% 0%	NA	NA	0%	0%	NA / NA	NA	More than 12[b]
Kuo[15] 2021	Single center	Retrospective	185	100%	100% / 0% / 0%	52.20[b]	78.92%	0%	0%	71.35% / NA	39.50[b]	35.50[b]
Chen[16] 2021	Single center	Prospective	129	70.54%	100% / 0% / 0%	34.30[b]	75.97%	0%	100%	NA / NA	56.20[b]	30.55[c,d]

Huang[17] 2021	Single center	Retrospective	301	100%	100% / 0% / 0%	51.70[b]	81.06%	0%	0%	NA / NA	42.15[b]	56.26[b]
Jeng[42] 2013	Single center	Prospective and Retrospective	95	100%	100% / 0% / 0%	52.10[c]	87.37%	41.05%	0%	88.42% / 58.95%	25.75[c]	12(max)[c]
Sohn[29] 2014	Single center	Retrospective	95	70.53%	100% / 0% /0%	48.83[c]	55.79%	46.32%	43.16%	NA / NA	22.00[c]	22.00[c]
Fong[44] 2015	Multicenter Single country	Retrospective	54	100%	94.44% / 5.56% / 0%	45.50[c]	62.96%	NA	100%	NA / NA	37.80[b]	31.10[c]
Jung[32] 2016	Single center	Prospective	113	71.68%	100% / 0% /0%	50.00[c]	61.95%	18.58%	39.82%	NA / NA	8.80[c]	12.50[c,d]
Lin[39] 2016	Single center	Retrospective	80	58.75%	100% / 0% / 0%	45.22[b]	81.25%	13.75%	41.25%	NA/ NA	36.00(fix)	12 (fix)
Wang[1] 2016	Multicenter Single country	Prospective	93	100%	100% / 0% /0%	46.41[b]	66.67%	0%	46.24	85.90% / 47.44%	39.80[b]	24.80[b]
Papatheodoridis[6] 2018	Multicenter International	Prospective	130	90.00%	41.54% / 58.46% / 0%	62.00[b]	69.23%	0%	0%	NA / NA	60.00[c]	15.00[c]
Wang[26] 2020	Multicenter Single country	RCT	46	100%	100% / 0% / 0%	47.30[b]	NA	0%	100%	NA / NA	NA	12 (fix)
Manolakopoulos[18] 2021	Multicenter Single country	Prospective	57	100%	0% / 100% / 0%	58.00[b]	70.18%	0%	0%	85.96% / 42.11%	90.00[b]	38[c]
Sonneveld[41] 2022	Multicenter International	Prospective and Retrospective	572	77.80%	79.90% / 15.91% / 4.19%	52.00[c]	68.18%	NA	16.61%	78.15% / 45.45%	73.74[c]	12 (max)[c,e]

Xie[27] 2021	Multicenter Single country	Prospective	139	82.73%	100% / 0% / 0%	36.00[c]	58.27%	0%	100%	84.17% / 54.68%	76.80[b]	12.00[c]
Xia[19] 2021	Single center	Prospective	135	54.82%	100% / 0% / 0%	36.11[b]	81.48%	0%	71%	NA / NA	At least 29.26[b,j]	31.20[c]
Liao[20] 2021	Single center	Prospective	122	58.20%	100% / 0% / 0%	34.00[c]	77.87%	0%	100%	74.59% /	56.40[c]	36.00[c]
Papatheodoridi[46] 2022	Multicenter International	Retrospective	325	89.85%	56.00% / 44.00% / 0%	56.00[b]	71.69%	0%	0%	NA/ NA	Causassian:60[c] Asian: 36[c]	60.08[c]
Broquetas[47] 2022	Single center	Prospective and Retrospective	52	98.06%	NA / 57.69% / NA	52.00[c]	75.00%	0%	0%	84.62% / 48.08%	98.04[c]	12 (max)
Hall[24] 2022	Multicenter	Prospective	110	90.00%	9.01% / 84.55% / 6.44%	56.00[c]	57.27%	0%	0%	80.91% / 40.00%	At least 24	24 (fix)
Chen[23] 2022	Single center	Prospective	316	100%	0% / 100% / 0%	40.41[b]	68.35%	0%	100%	NA / NA	46.05[b]	42[c]
Peng[45] 2022	Single center	Prospective	1309	100%	0% / 100% / 0%	53.40[c]	83.58%	40.87%	0%	82.82% / NA	34[c]	51[c]
Papatheodoridi[22] 2022	Multicenter	Prospective	57	100%	100% / 0% / 0%	60.00[c]	64.91%	0%	0%	NA / NA	98.40[b]	38.00[b]

ALT: alanine aminotransferase , EHR: electronic health record, DNA: deoxyribonucleic acid, EOT: end of treatment, ETV: entecavir, HBeAg: hepatitis B envelope antigen, HBsAg: hepatitis B surface antigen, NA: not available, Nuc: nucleo(s)tide analogue, RCT: randomized controlled trial, SAE: sever acute exacerbation, defined by ALT>5X ULN and bilirubin > 2mg/dL or prolonged prothrombin time >3 seconds before antiviral agents before antiviral agents, TDF: tenofovir disoproxil fumarate.

a: data from the off-Nuc cohort, b: mean, c: median, d: the number was estimated according to survival curve or the content of the manuscript, e: 88% patients completed the 12 months follow-up, f: total 124 patients stopped TDF, and we extracted the data from 106 patients with negative hepatitis B e-antigen in the end of TDF treatment, g: the data were from the total 124 patients, including some positive hepatitis B e-antigen population in the end of TDF treatment, because the data of the 106 patients with negative hepatitis B e-antigen in the end of treatment were

unavailable, h: data from subgroup of patients exclusively receiving ETV or TDF, i: duration of viral remission under antivirals, not the total duration of treatment, i: the data was retracted from single center cohort from a multicenter study, j: this is consolidation duration instead of treatment duration, which was not provided in the manuscript.

Table S6: Summary of the Stopping and Resuming Nuc Criteria of the 50 Enrolled Studies

Author Year	Stopping Nuc criteria	Resuming Nuc criteria
Shouval[25] 2009	HBV DNA < 0.7 mEq/mL and ALT < 1.25X ULN at week 48	NA
Buti[40] 2019	TDF more than 8 years and HBV viral load < 29 IU/mL	NA
Liem[11] 2019	EASL 2017[a]	HBeAg seroreversion; HBV DNA >2000 IU/mL and ALT >600 IU/mL (any visit); HBV DNA >2000 IU/mL and ALT >5X ULN (two consecutive visits); HBV DNA >2000 IU/mL and ALT >200 IU/mL but <600 IU/mL for >6–8 weeks; HBV DNA >20 000 IU/mL on two consecutive visits at least 4 weeks apart
Wong[34] 2020	NA	NA
Seto[50] 2021	EASL 2017[a]	HBV DNA ≥ 2000 IU/mL two times within one week
Lai[21] 2021	Taiwan reimbursement[a]	Taiwan reimbursement[d]
Hsu[35] 2022	Taiwan reimbursement[a]	Taiwan reimbursement[d]
Hirode[36] 2022	As different institutions[b]	As different institutions[b]
Liu[37] 2022	APASL 2012[a]	Taiwan reimbursement[d]
Seto[28] 2015	APASL 2012[a]	HBV DNA >2000 IU/mL

Chen[30] 2015	APASL 2012[a]	No description
Chen[31] 2015	APASL 2012[a]	HBV-DNA > 2000 IU/mL
Jeng[38] 2016	Taiwan reimbursement[a]	Taiwan reimbursement[d]
Chien[2] 2017	Taiwan reimbursement[a]	Taiwan reimbursement[d]
Hsu[3] 2017	Taiwan reimbursement[a]	Taiwan reimbursement[d]
Yao[48] 2017	APASL 2012[a,c]	Taiwan reimbursement[d]
Chen[4] 2018	APASL 2012[a]	Taiwan reimbursement[d]
Chen[43] 2018	APASL 2012[a]	NA
Jeng[33] 2018	APASL 2008[a]	Taiwan reimbursement[d]
Papatheodoridis[5] 2018	Nuc ≥ 4 years and monotherapy with ETV or TDF f ≥2 years; HBV DNA undetectability ≥3 years (similar with EASL 2017)	ALT >10×ULN; ALT >5×ULN & total bilirubin >2 mg/dL (the same visit); ALT >3×ULN and HBV DNA >100,000 IU/mL (the same visit); ALT >ULN and HBV DNA >2,000 IU/mL (three sequential visits).
Su[7] 2018	APASL 2012[a]	Taiwan reimbursement[d]
Chen[8] 2019	APASL 2012[a]	Taiwan reimbursement[d]
Kuo[9]	APASL 2012[a]	Taiwan reimbursement[d]

2019		
Ma[10] 2019	APASL 2012[a]	Taiwan reimbursement[d]
Chen[12] 2020	APASL 2012[a]	Taiwan reimbursement[d]
Chiu[13] 2020	APASL 2012[a]	Taiwan reimbursement[d]
Tseng[49] 2020	APASL 2012[a]	Taiwan reimbursement[d]
Fang[14] 2021	APASL 2012[a]	Taiwan reimbursement[d]
Kuo[15] 2021	APASL 2012[a]	Taiwan reimbursement[d]
Chen[16] 2021	APASL 2102[a]	HBV DNA >2000 IU/mL and ALT >2X ULN
Huang[17] 2021	APASL 2012[a]	Taiwan reimbursement[d]
Jeng[42] 2013	APASL 2008[a]	NA
Sohn[29] 2014	APASL 2008[a]	HBV DNA >60 IU/mL
Fong[44] 2015	HBeAg seroconversion with undetectable HBV DNA ≥ 1 month	NA
Jung[32] 2016	APASL 2012[a]	HBV DNA >2000 IU/mL
Lin[39]	Taiwan reimbursement[a]	NA

2016		
Wang[1] 2016	APASL 2012[a] or Taiwan reimbursement	HBV DNA >2000 IU/mL and ALT >2X ULN (but not clearly defined)
Papatheodoridis[6] 2018	Undetectable HBV DNA ≥24 months (Similar with APASL 2008)[a]	Greek centers: ALT >10X ULN, or ALT >5X ULN and total bilirubin >2 mg/dL, or ALT>3X ULN and HBV DNA >100,000 IU/mL, or ALT >ULN and HBV DNA >2000 IU/ mL on three sequential occasions. Taiwan center :ALT >2X ULN twice 3 months apart and HBV DNA >2,000 IU/mL, or total bilirubin >2 mg/dL, or PT prolongation ≥3 seconds.
Wang[26] 2020	HBeAg seroconversion and consolidation therapy for 12-18 months (similar with APASL 2012)[a]	NA
Manolakopoulos[18] 2021	Nuc >4 years with ETV or TDF >2 years; Undetectable HBV DNA >3 years; No advanced fibrosis; No history of decompensation (similar with EASL 2017)	ALT >10X ULN, or ALT >5X ULN and total bilirubin > 2mg/dL, or ALT > 3X ULN and HBV DNA >100,000 IU/mL, or ALT >ULN and HBV DNA > 2000 IU/mL on three sequential
Sonneveld[41] 2022	As different institutions	As different institutions
Xie[27] 2021	Undetectable serum HBV DNA, normal ALT levels, HBeAg seroconversion for ≥ 3 years, and Nuc ≥4 years (as Chinese guidelines)	HBV DNA >2000 IU/mL and ALT >2X ULN
Xia[19] 2021	HBeAg(+) patients: consolidation ≥12months; HBeAg(-) patients: consolidation ≥18 months. (similar with APASL 2012)	HBV DNA >2000 IU/mL and ALT >2X ULN
Liao[20] 2021	APASL 2012[a]	HBV DNA >2000 IU/mL and ALT >2X ULN
Papatheodoridi[46] 2022	Caucasian: EASL 2017[a] Asian: APASL 2012[a]	Greek center: ALT >10X ULN, or ALT >5 X ULN, and total bilirubin >2 mg/dL or ALT >3X ULN and HBV DNA >100,000 IU/mL, or ALT > ULN and HBV DNA >2000 IU/mL on three sequential occasions.

		Taiwan center : Taiwan reimbursement.[d]
Broquetas[47] 2022	EASL 2017[a]	Any of the following criteria: severe flares (ALT >10X ULN in two consecutive blood test for 2 weeks), moderate flares (ALT 5–10X ULN in two consecutive blood test for 4 weeks) or mild persistent flares (ALT 2–5X ULN and DNA >2000 IU/mL persisting for more than 6 months).
Hall[24] 2022	Continuous NA therapy for at least 2 years with HBV DNA levels <20 IU/mL on three consecutive occasions at least 6 months apart over 18 months	(1)HBV DNA > 2000 IU/ ml and serum ALT >5× ULN for ≥16 weeks or ALT >10× ULN for ≥8 weeks; (2)Clinical evidence of hepatic decompensation defined by INR ≥1.5 or bilirubin >2X ULN or ascites or hepatic encephalopathy; or (3) investigator discretion
Chen[23] 2022	Taiwan reimbursement[a]	Taiwan reimbursement[d]
Peng[45] 2022	APASL 2008[a]	Taiwan reimbursement[d]
Papatheodoridi[22] 2022	Nuc ≥ 4 years and monotherapy with ETV or TDF f ≥2 years; HBV DNA undetectability ≥3 years (similar with EASL 2017)	ALT >10×ULN; ALT >5×ULN & total bilirubin >2 mg/dL (the same visit); ALT >3×ULN and HBV DNA >100,000 IU/mL (the same visit); ALT >ULN and HBV DNA >2,000 IU/mL (three sequential visits); or patients' and physicians' decisions in case of HBV DNA >20,000 IU/mL

ALT: alanine aminotransferase, EASL: European Association for the Study of the Liver, ETV: entecavir, DNA: deoxyribonucleic acid, HBeAg: hepatitis B e-antigen, HBV: hepatitis B virus, INR: international normalized ratio, NA: not available, Nuc: nucleo(s)tide analogue, PT: prothrombin time, TDF: tenofovir disoproxil fumarate, ULN: upper limit of normal.

a: supplementary table 9, b: supplementary table 10, c: for ETV patients, d: ALT >2X ULN with 3 months apart and HBV DNA >2000 IU/mL or total bilirubin >2 mg/dL, or prolongation of PT ≥3 seconds.

Table S7: Summary of the Safety Outcomes of the 50 Enrolled Studies

Author Year	ALT > 5X ULN N (%)	ALT >10X ULN N (%)	Severe hepatitis flares or decompensation[a] N (%)	Hepatitis Flare-related death / Liver transplantation N (%)
Shouval[25] 2009	NA Follow-up: fixed 6 months	24 (9.34%) Follow-up: fixed 6 months	0 (0%)[b,c] Follow-up: fixed 6 months	0 (0%) Follow-up: fixed 6 months
Buti[40] 2019	29 (27.36%) Follow-up: up to 6 months	Incomplete data	3 (2.83%)[d] Follow-up: up to 6 months	0 (0%) Follow-up: up to 6 months
Liem[11] 2019	22 (48.89%) Follow-up:fixed 18 months	NA	1 (2.22%)[e] Follow-up: fixed 18 months	0% Follow-up: fixed 18 months
Wong[34] 2020	85 (7.90%)	39 (3.63%)	Serum bilirubin >2X ULN: 114 (10.60%); Serum bilirubin >5X ULN: 19 (1.77%); Decompensation: 7 (0.65%)[p]	NA
Seto[50] 2021	NA	4 (3.51%) Follow-up: fixed 12 months	NA	0 (0%) Follow-up: fixed 12 months
Lai[21] 2021 Prior SAE cohort	NA	23 (29.49%)	11 (14.10%)[f]; Cumulative incidence: 1Y: 8.83%; 2Y: 14.12%; 3Y: 17.42%; 4Y: 17.42%; 5Y: 25.91%	1 (1.28%)
Lai[21] 2021 Non-Prior SAE cohort	NA	28 (17.95%)	1 (0.64%)[f]; Cumulative incidence: 1Y: 0%; 2Y: 1.04%; 3Y: 1.04%; 4Y: 1.04%; 5Y: 1.04%	1 (0.64%)
Hsu[35][r] 2022	NA	NA	24 (3.61%)[g,r]	2 (0.30%)[r]

Hirode[36] 2022	359 (23.13%), Cumulative incidence: 1Y:18.6%, 2Y:30.8%	NA	19 (1.22%)[h]; Annual incidence: 0.05%	4 (0.26%)
Liu[37] 2022	516 (41.82%), Cumulative incidence: 1Y:29%, 2Y:42%	NA	Bilirubin ≥2mg/dL: 73 (5.92%) Decompensation: 13 (1.05%)[k]	5 (0.41%)
Seto[28] 2015	NA	NA	0 (0%)[i]	0 (0%)
Chen[30] 2015	NA	NA	7 (2.78%)[j]	2 (0.79%)
Chen[31] 2015	NA	NA	7 (4.27%)(at least)[s]	Incomplete data[u]
Jeng[38] 2016	31 (36.47%)	22 (25.88%)	Bilirubin ≥2 mg/dL:9 (10.60%) Decompensation:2 (2.35%)[k]	0 (0%)
Chien[2] 2017	NA	NA	0 (0%)[f]	0 (0%)
Hsu[3] 2017	NA	NA	4 (3.01%)[f]	0 (0%)
Yao[48] 2017	NA	NA	0 (0%)[j]	0 (0%)
Chen[4] 2018	44 (30.77%)	26 (18.18%)	2 (1.40%)[j]	0 (0%)
Chen[43] 2018	NA	NA	5 (1.22%)[j]	1 (0.24%)
Jeng[33] 2018	NA	NA	7 (1.01%)[k]; Cumulative incidence: 5Y: overall: 1% (non-cirrhotic patients: 0%;	3 (0.43%)

			cirrhotic patients: 2.95%); Annual incidence: 0.28%	
Papatheodoridis[5] 2018	14 (24.56%)	4 (7.02%)	0 (0%)[l]	0 (0%)
Su[7] 2018	NA	NA	0 (0%)[f]	0 (0%)
Chen[8] 2019	NA	NA	2 (0.86%)[j]	0 (0%)
Kuo[9] 2019	156 (30.77%)	96 (18.93%)	APASL definition: 8 (1.58%)[m]	0 (0%)
Ma[10] 2019	128 (23.93%)	66 (12.34%)	Bilirubin>2mg/dL: 44 (8.22%); Bilirubin > 3mg/dL: 10 (1.87%); APASL definition: 7 (1.31%)[m]	1 (0.19%)
Chen[12] 2020	NA	NA	APASL definition: 2 (0.80%)[m]	0 (0%)
Chiu[13] 2020	NA	NA	5 (1.09%)[n]	NA
Tseng[49] 2020	NA	NA	2 (1.48%)[f]	1 (0.74%)
Fang[14] 2021	NA	NA	4 (0.84%)[f]	1 (0.21%)
Kuo[15] 2021	NA	NA	2 (1.08%)[f]	1 (0.54%)
Chen[16] 2021	15 (11.63%)	NA	0 (0%)[o]	0 (0%)
Huang[17]	NA	NA	2 (0.66%)[f]	0 (0%)

2021				
Jeng[42] 2013	NA	NA	1 (1.05%)[c]	0 (0%)
Sohn[29] 2014	NA	NA	0 (0%)[c]	0 (0%)
Fong[44] 2015	NA	NA	0 (0%)[c]	0 (0%)
Jung[32] 2016	NA	NA	0 (0%)[c]	0 (0%)
Lin[39] 2016	NA	NA	0 (0%)[c]	0 (0%)
Wang[1] 2016	NA	NA	1(1.08%)[c.t]	1(1.08%)
Papatheodoridis[6] 2018	NA	NA	0 (0%)[c]	0 (0%)
Wang[26] 2020	NA	NA	0 (0%)[c]	0 (0%)
Manolakopoulos[18] 2021	11 (19.30%)	10 (17.54%)	1 (1.75%)[t.v]	0 (0%)
Sonneveld[41] 2022	NA	NA	2 (0.35%)[q]	0 (0%)
Xie[27] 2021	NA	NA	0 (0%)[c]	0 (0%)
Xia[19] 2021	NA	NA	0 (0%)[c]	0 (0%)
Liao[20]	NA	NA	0 (0%)[c]	0 (0%)

2021				
Papatheodoridi[46] 2022	NA	NA	NA	0 (0%)
Broquetas[47] 2022	NA	3 (5.77%)	NA	0 (0%)
Hall[24] 2022	35 (31.82%)	24 (21.82%)	0 (0%)[t]	0 (0%)
Chen[23] 2022	NA	59 (18.67%)	5 (1.58%)[u]	1 (0.32%)
Peng[45] 2022	NA	245 (18.72%)	NA[v]	5 (0.48%)
Papatheodoridi[22] 2022	NA	NA	0 (0%)[l]	0 (0%)

ALT: alanine aminotransferase, APASL: Asian Pacific Association for the Study of the Liver, NA: not available, INR: international normalized ratio; Nuc: nucleo(s)tide analogue, SAE: severe acute exacerbation: ALT>5X ULN and bilirubin > 2mg/dL or prolonged prothrombin time >3 seconds before antiviral agents, ULN: upper limit of normal, WHO: World Health Organization.

a: each study has its own definition of hepatitis flares or decompensation. For studies without their own definition, we extracted the data about bilirubin at least more than 2mg/dL, coagulopathy, or mention of clinical decompensation such as ascites, jaundice, or encephalopathy from the studies, b: patients of off-Nuc cohort, c: without clear definition, d: this study reported the adverse events according to grading systemic without the definition of decompensation. We extracted the data according to grade 2 hyperbilirubinemia (>1.5X ULN), e: this study reported the adverse events according to WHO grading systemic without the definition of decompensation. We extracted the data according to grade 3 hyperbilirubinemia (>66 umol/L), f: defined by bilirubin >2mg/dL or prolonged prothrombin time >3 seconds, g: defined by ALT >5X ULN and bilirubin >2mg/dL, h: defined by bilirubin level ≥2 mg/d, increased INR, appearance of clinical jaundice, onset of ascites, variceal bleeding, or hepatic encephalopathy, i: defined by Child-Pugh score B or C, j: defined by bilirubin ≥3 mg/ mL and prolonged prothrombin time ≥3 seconds, k: defined by jaundice and INR > 1.5 and/or occurrence of ascites/encephalopathy, l: defined by the study's statement "no patient experienced jaundice or bilirubin >2 mg/dL, or developed any sign of liver decompensation", m: defined by APASL definition of decompensation

“ bilirubin >3.5 mg/dL [>2.5 X ULN(1.4 mg/dL)] and prolonged PT >3 seconds, or INR >1.5”, n: defined by INR >1.5, o: defined by bilirubin >2.5X ULN, p: defined by clinical presentation of decompensation, q: defined by study’s statement “two patients developed jaundice “, r: subgroup of patients exclusively receiving ETV or TDF, s: the study reported “7 patients with varices bleeding and 6 patients with ascites” but didn’t clarify whether there were overlapping patients, t: at least one patient with decompensation mentioned in the manuscript, u: study reported one patient died but didn’t clarify whether it was flare-related liver death, v: defined by bilirubin >2mg/dL, t: defined by INR ≥1.5 or serum bilirubin level >2× ULN or ascites or hepatic encephalopathy, u: bilirubin >3.5 mg/dL (>2.5 times the ULN [1.4 mg/dL]) or an INR >1.5, v: bilirubin > 2 mg/dL and INR > 1.5 and/or occurrence of ascites/encephalopathy

Table S8: Summary of the Design for the 10 Studies Using Safety Outcomes as the Prespecified Events

Author Year	Study Setting	Cirrhosis enrollment	Number	Follow-up (months)	Main Prespecified Safety Outcomes	Off-Nuc principle	Retreatment principle
Shouval[25] 2009	Prospective Multicenter	No	257	6 (fix)	ALT >10X ULN	HBV DNA < 0.7 mEq/mL and ALT < 1.25 X ULN at Week 48	NA
Buti[40] 2019	Prospective Multicenter	Yes	106[d]	6 (fix)	ALT >10X ULN and Liver toxicity Grading System	TDF ≥8 years and HBV DNA < 29 IU/mL	NA
Liem[11] 2019	RCT Single Center	No	45	18 (fix)	WHO Grading system	EASL 2017[a]	As trial protocol[h]
Wong[34] 2020	Retrospective / EHR Multicenter	Yes	1076	44.30[b]	ALT >10X ULN	NA	NA
Seto[50] 2021	Prospective Multicenter	No	114 (All with HBsAg < 200 IU/mL)	12 (max)	ALT >10X ULN	EASL 2017[a]	DNA ≥ 2000 IU/mL
Lai[21] 2021	Retrospective	No	SAE group: 78 Non SAE group: 156	19.56[c] 24.87[c]	ALT >10X ULN	Taiwan reimbursement[a]	Taiwan reimbursement[f]
Hsu[35] 2021	Retrospective / EHR Single Center	No	665[e]	39.60[c,e]	ALT >5X ULN and Bilirubin >2mg/dL	Taiwan reimbursement[a]	Taiwan reimbursement[f]
Hirode[36] 2022	Prospective Multicenter	Yes	1552	18.40[c]	Bilirubin >2mg/dL, or coagulopathy, or clinical decompensation		As different institutions[g]

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Liu[37] 2022	Prospective Single Center	Yes	1234	50.25[c]	Bilirubin ≥2mg/dL, and coagulopathy, or clinical decompensation	APASL 2012[a]	Taiwan reimbursement[f]
Hall[24] 2022	Prospective Multicenter	No	110	56.00[c]	ALT >10X ULN, and decompensation event	Similar with APASL 2012[a]	(1)HBV DNA > 2000 IU/ ml and serum ALT >5× ULN for ≥16 weeks or ALT >10× ULN for ≥8 weeks; (2)Clinical evidence of hepatic decompensation (3) investigator discretion

ALT: alanine aminotransferase, APASL: Asian Pacific Association for the Study of the Liver, DNA: deoxyribonucleic acid, EHR: electronic health record, NA: not available, RCT: randomized controlled trial, TDF: tenofovir disoproxil fumarate, ULN: upper limit of normal, WHO: World Health Organization

a: supplementary table 9, b: mean, c: median, d: total 124 patients stopped TDF, and we extracted the data from 106 patients with negative hepatitis B e-antigen in the end of TDF treatment, ,e: data from subgroup of patients exclusively receiving ETV or TDF, f: ALT >2X ULN with 3 months apart and HBV DNA >2000 IU/mL or total bilirubin >2 mg/dL, or prolongation of PT ≥3 seconds, g: supplementary table 9, h: any of the following criteria: HBeAg seroreversion, HBV DNA >2000 IU/mL and ALT >600 IU/mL at any visit, HBV DNA >2000 IU/mL and ALT >5x ULN on two consecutive visits, HBV DNA >2000 IU/mL and ALT >200 IU/mL but <600 IU/mL for >6–8 weeks, HBV DNA >20 000 IU/mL on two consecutive visits at least 4 weeks apart.

Table S9: Off-Nuc Criteria for CHB Patients without HBsAg Loss according to International Guideline[a] or Local Regulation

	APASL 2008[51]	APASL 2012[52]	APASL 2016[53]	EASL 2017[54]	AASLD 2018[55]
HBeAg(+)	• HBeAg seroconversion • Consolidation with undetectable HBV DNA ≥ 6 months	• HBeAg seroconversion • Consolidation with undetectable HBV DNA ≥12 months	• Non-cirrhosis • HBeAg seroconversion • Consolidation with undetectable HBV DNA ≥ 12 months (36 months was preferred)	• Non-cirrhosis • HBeAg seroconversion • Consolidation with undetectable HBV DNA ≥ 12 months • Evidence: II-2; Recommendation: Weaker	• Non-cirrhosis • HBeAg seroconversion • Consolidation with normal ALT and undetectable HBV DNA ≥ 12 months • Evidence: low; Recommendation: conditional
HBeAg(-)	• Consolidation with undetectable HBV DNA ≥12 months	• Treatment duration ≥24 months • Consolidation with undetectable HBV DNA ≥12 months	• Non-cirrhosis • Treatment duration ≥24 months • Consolidation with undetectable HBV DNA ≥12 months	• Non-cirrhosis • Undetectable HBV DNA ≥ 36 months • Evidence: II-2; Recommendation: Weaker	• Indefinite treatment duration • Evidence: low; Recommendation: conditional
	Taiwan reimbursement[3][b]				
HBeAg(+)	• Non-cirrhosis • HBeAg loss • Consolidation therapy for 12 months				
HBeAg(-)	• Treatment ≥ 2 years with Consolidation therapy for 12 months • Maximum duration of 3 years				

AASLD: American Association for the Study of Liver Diseases, ALT: alanine aminotransferase, APASL: Asian Pacific Association for the Study of the Liver, CHB: chronic hepatitis B, EASL: European Association for the Study of the Liver, HBeAg: hepatitis B e-antigen, HBsAg: hepatitis B virus s-antigen, WHO: World Health Organization.
a: in the condition of highly motivated patients who could participate close post-Nuc monitoring, b: this is general principle not for selected population.

Table S10: Off-Nuc and Retreatment Criteria of Each Institutions Participating in RETRACT-B study[36]

Center country	Number of centers	Study design	NA stopping criteria	Retreatment criteria
Belgium	1 *	Cohort	HBeAg negative for at least 6 months, at the discretion of the treating physician, patient's own initiative	Belgian reimbursement criteria, at the discretion of the treating physician
Germany	1	Cohort, Trial	HBeAg negative with at least 42 months of undetectable HBV DNA	At the discretion of the treating physician
Greece	1	Trial	HBeAg negative with at least 36 months of undetectable HBV DNA	Virological relapse, combined relapse, ALT >10x ULN, ALT >3x ULN and HBV DNA >100,000 IU/mL at the same visit, ALT >ULN and HBV DNA >2,000 IU/mL on three sequential visits, patients' and physicians' decisions in case of HBV DNA >20,000 IU/mL
Netherlands	1	Cohort, Trial	HBeAg negative with at least 12 months of undetectable HBV DNA, patient's own initiative	At the discretion of the treating physician, patient's own initiative
Spain	1	Cohort, Trial	HBeAg negative with at least 36 months of undetectable HBV DNA	At the discretion of the treating physician
Hong Kong	3	Cohort, Trial	APASL guidelines	Virological relapse regardless of ALT level
Taiwan	4	Cohort, Trial	APASL guidelines, Taiwan's national health plan, patient's own initiative	Taiwan's national health plan, hyperbilirubinemia (serum total bilirubin >2 mg/dL), coagulopathy (prothrombin time prolongation >3 seconds), combined relapse, at the discretion of the treating physician, patient's own initiative
Canada	1	Cohort, Trial	HBeAg negative with at least 12 months of undetectable HBV DNA	HBeAg seroreversion, HBV DNA >2000 IU/mL and ALT >600 IU/mL at any visit, HBV DNA >2000 IU/mL and ALT >5x ULN on two consecutive visits, HBV DNA >2000 IU/mL and ALT >200 IU/mL but <600 IU/mL for >6–8 weeks, HBV DNA >20 000 IU/mL on two consecutive visits at least 4 weeks apart, at the discretion of the treating physician

* Data was centralized at one center however it was collected from 18 centers across Belgium.

APASL, Asia-Pacific Association for the Study of the Liver; ULN, Upper limit of normal.

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Table S11: Pooled Safety Outcomes of Selected Studies[a]

	Study Number	Patient Number	Pooled Estimate (95% CI)	Heterogeneity	
				I^2	p
Severe Hepatitis Flares or Decompensation from Selected Studies					
Overall population	15[1, 5, 7, 10, 11, 18, 19, 21, 24, 29, 32, 34, 35, 37, 44]	4525	1.21% (0.70-2.08%)	54%	< 0.01
Non-cirrhotic population	14[1, 5, 7, 10, 11, 18, 19, 21, 24, 29, 32, 34, 35, 37]	3731	0.89% (0.44-1.79%)	50%	0.02
Cirrhotic population	5[29, 32, 34, 35, 37]	744	3.63% (1.77-7.29%)	42%	0.14
Severe Hepatitis Flares or Decompensation from Selected Studies with Clear Definition					
Overall population	9[5, 7, 10, 11, 21, 24, 34, 35, 37]	3978	1.27% (0.68-2.37%)	70%	< 0.01
Non-cirrhotic population	9[5, 7, 10, 11, 21, 24, 34, 35, 37]	3299	0.85% (0.36-2.02%)	66%	< 0.01
Cirrhotic population	3[34, 35, 37]	679	3.88% (1.86-7.91%)	65%	0.06
Hepatitis Flare-related Death, or Liver Transplantation from Selected Studies					
Overall population	14[1, 5, 7, 10, 11, 18, 19, 21, 24, 29, 32, 35, 37, 44]	3499	0.37% (0.20-0.67%)	0%	1.00
Non-cirrhotic population	13[1, 5, 7, 10, 11, 18, 19, 21, 24, 29, 32, 35, 37]	2744	0.30% (0.12-0.73%)	0%	0.98
Cirrhotic population	4[29, 32, 35, 37]	655	0.98% (0.45-2.14%)	0%	0.96

Estimated risks, generated by random-effects model and the inverse variance method, are expressed by proportions with 95% confidence internal. Between-study heterogeneity are represented by I^2 statistics.

a: criteria of study selection for pooling aggregate data: (1) no overlapping populations, (2) follow-up duration >12 months, (3) not for special populations such as studies exclusively composed of patients with prior severe acute exacerbation or low serum HBsAg level.

CI: confidence interval, HBsAg: hepatitis B virus surface antigen.

Table S12: The Definitions of Liver Cirrhosis or Advanced Fibrosis of the 15 Studies Selected for Risk Estimation and Two Multinational Studies Containing Patient Populations Overlapped in Other Published Studies

Author Year	Patient number	Cirrhosis (%)	Definition of cirrhosis or advanced fibrosis
Liem[11] 2019	45	0%	History within 5 years, liver stiffness measurement (FibroScan) >9kPa and/or Fibrotest >0.48
Wong[34] 2020	1076	8.27%	ICD-9-CM diagnosis codes for cirrhosis and its related complications
Lai[21] 2021 Non-prior SAE group	156	0%	Image studies and supplementary clinical/laboratory features
Hsu[35] 2022	665	14.29%	Image studies and evidence of portal hypertension (e.g., splenomegaly or varices)
Liu[37] 2022	1234	40.11%	Histologic or repeated ultrasonography consistent with cirrhosis, supplemented with the presence of varices, splenomegaly and/or thrombocytopenia
Papatheodoridis[5] 2018	57	0%	Histology (Ishak staging score >4) or liver stiffness >10 kPa at a reliable liver elastography
Su[7] 2018	100	0%	Ultrasound findings
Ma[10] 2019	535	0%	Histology , a Fibrosis-4 (FIB-4) or liver stiffness measurement, or combined repeated ultrasound findings and with clinical features such as splenomegaly, thrombocytopenia, ascites, or gastroesophageal varices at baseline.
Sohn[29] 2014	95	46.32%	Histology or a combination of clinical findings indicating the presence of portal hypertension
Fong[44] 2015	54	NA	NA
Jung[32] 2016	113	18.58%	Histology or the following clinical criteria: (1) platelet count < 100,000/lL and ultrasonographic findings suggestive of cirrhosis, including a blunted, nodular liver edge accompanied by splenomegaly (12 cm); or (2) evidence of portal hypertension-related complications
Wang[1] 2016	93	0%	NA[a]
Manolakopoulos[18] 2021	57	0%	Histology (Ishak staging score >4) or liver stiffness >10 kPa at a reliable liver elastography

Xia[19] 2021	135	0%	Elastography (Fibroscan; Echosens).
Hall[24] 2022	110	0%	Histology (METAVIR >F4) or liver stiffness > 9.5 kPa
RETRACT-B Study Hirode[36] 2022	1552	11.86%	NA
CREATE Study Sonneveld[41] 2022	572	NA	NA

NA: not available.

a: The definition was not mentioned in the context, but should be mainly based on reimbursement criteria in Taiwan (image finding of cirrhosis in combination with splenomegaly or varices).

Reference(Author names in bold designate shared co-first authorship

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